The documentation and process conversion measures necessary to comply with this document shall be completed by 1 August 2005.

INCH-POUND

MIL-PRF-19500/368G
1 May 2005
SUPERSEDING
MIL-PRF-19500/368F
8 July 2002

* PERFORMANCE SPECIFICATION SHEET

* SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES: 2N3439, 2N3439L, 2N3439UA, 2N3439U4, 2N3440, 2N3440L, 2N3440UA,
AND 2N3440U4, JAN, JANTX, JANTXV, JANS, JANHCA, JANKCA, JANHCB, AND JANKCB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low-power, high voltage transistors. Four levels of product assurance are provided for each encapsulated device types as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die.

* 1.2 Physical dimensions. See figure 1 (similar to TO-5 and TO-39), figure 2 (JANHCA and JANKCA (A versions)), figure 3 (JANHCB and JANKCB (B versions)), figure 4 (2N3439UA and 2N3440UA surface mount versions), and figure 5 (2N3439U4 and 2N3439U4) versions).

* 1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T (1) T_A = +25°C	P_T (2) T_C = +25°C	P_T (2) T_{SP} = +25°C	$R_{\theta JA}$ (3)	$R_{\theta JC}$ (3)	$R_{\theta JSP}$ (3)	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_{STG} and T_J
	W	W	W	°C/W	°C/W	°C/W	V dc	V dc	V dc	A dc	°C
2N3439,	0.8	5.0	N/A	175	30	N/A	450	7	350	1.0	-65 to
2N3439L,	0.8	5.0	N/A	175	30	N/A	450	7	350	1.0	+200
2N3439UA	0.8	N/A	2.0	175	N/A	70	450	7	350	1.0	
2N3439U4	N/A	20.0	N/A	N/A	8	N/A	450	7	350	1.0	
2N3440,	0.8	5.0	N/A	175	30	N/A	300	7	250	1.0	-65 to
2N3440L	0.8	5.0	N/A	175	30	N/A	300	7	250	1.0	+200
2N3440UA	0.8	N/A	2.0	175	N/A	70	300	7	250	1.0	
2N3440U4	N/A	20.0	N/A	N/A	8	N/A	300	7	250	1.0	

(1) For derating, see figure 6.
(2) For derating, see figures 7, 8, and 9.
(3) For thermal impedance curves see figures 10, 11, 12, and 13.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact in formation can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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1.4 Primary electrical characteristics unless otherwise specified T_A = +25°C.

	h_{FE2} (1) V_{CE} = 10 V dc I_C = 2 mA dc	h_{FE1} (1) V_{CE} = 10 V dc I_C = 20 mA dc	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 10 mA dc f = 5 MHz	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	$V_{BE(sat)}$ (1) I_C = 50 mA dc I_B = 4 mA dc	$V_{CE(sat)}$ I_C = 50 mA dc I_B = 4 mA dc
				pF	V dc	V dc
Min	30	40	3			
Max		160	15	10	1.3	0.5

(1) Pulsed, (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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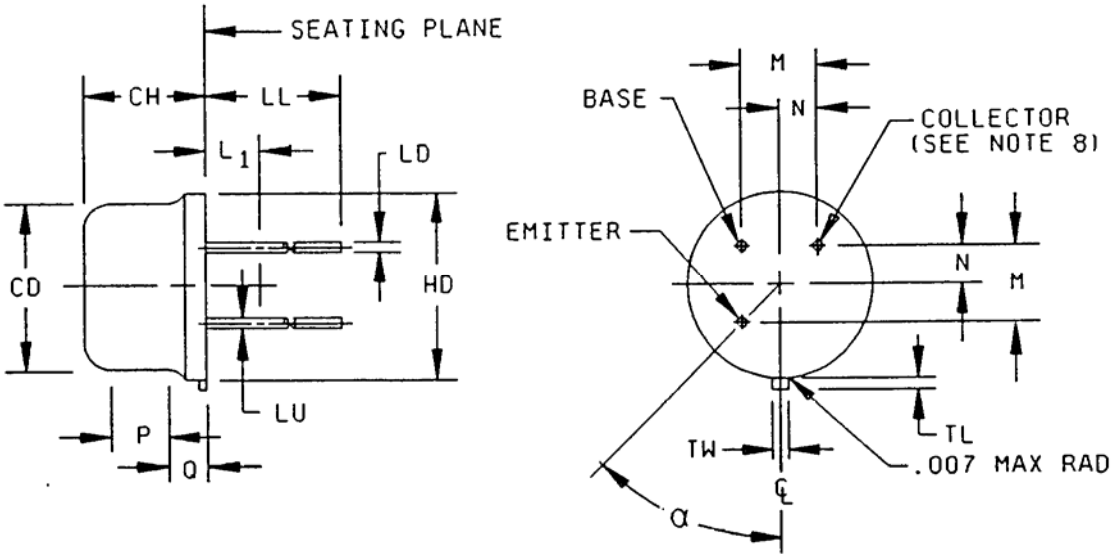


FIGURE 1. Physical dimensions (similar to TO-5 and TO-39).

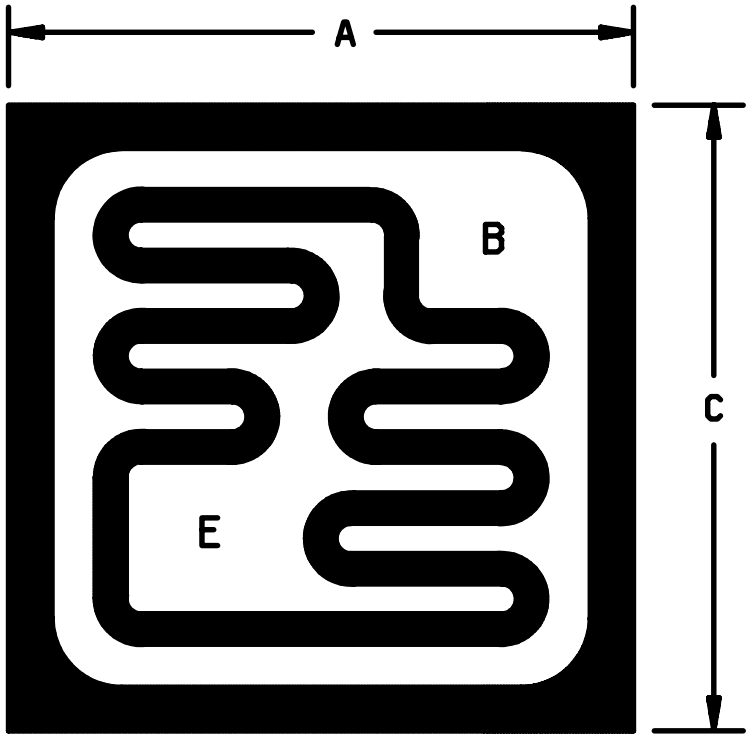
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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LD	.016	.021	0.41	0.53	3, 10
LL	See notes 10, 12, and 13				
L1		.050		1.27	11
LU	.016	.019	0.41	0.48	4, 10
M	.1414 Nom		3.591 Nom		7
N	.0707 Nom		1.796 Nom		7
P	.100		2.54		5
Q					6
TL	.029	.045	0.74	1.14	9
TW	.028	.034	0.71	0.86	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .50 inch (1.27 mm) from the seating plane.
5. Variations on dimension CD in this zone shall not exceed .010 inch (0.25 mm).
6. Outline in this zone is not controlled.
7. When measured in a gauging plane .054 +.001, -.000 inch (1.37 +0.03, - 0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
8. The collector shall be electrically connected to the case.
9. Measured from the maximum diameter of the actual device.
10. All three leads (see 3.4.1).
11. Diameter of leads in this zone is not controlled.
12. For transistor types 2N3439 and 2N3440, L = .500 inch (12.70 mm) minimum, and .750 inch (19.05 mm) maximum.
13. For transistor types 2N3439L and 2N3440L, L = 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum.
14. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions - Continued.

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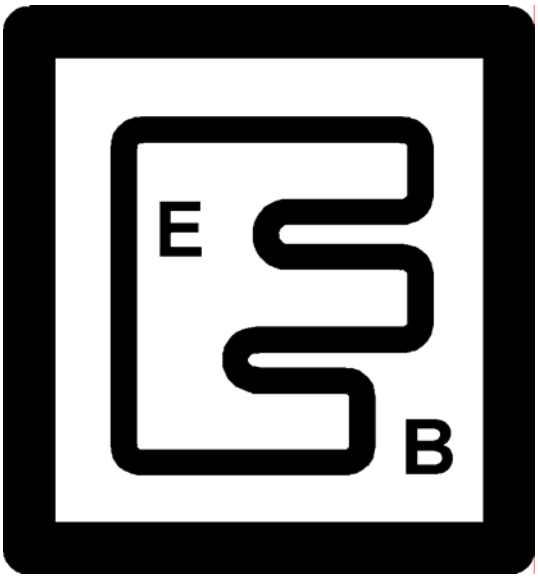


Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	.038	.044	0.97	1.12
C	.038	.044	0.97	1.12

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die thickness are, .006 inch (0.15 mm) to .012 inch (0.30 mm).
   Metallization, top: Al = 17,500 Å minimum, 20,000 Å nominal.
   Back: Au = 2,500 Å minimum, 3,000 Å nominal.
   Bonding pad: B = .004 inch (0.10 mm) by .005 inch (0.13 mm).
   E = .004 inch (0.10 mm) by .005 inch (0.13 mm).
4. Backside is collector.
5. Requirements in accordance with MIL-PRF-19500 are performed in a TO-5 package.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 2. Physical dimensions JANHCA and JANKCA (die) A versions.

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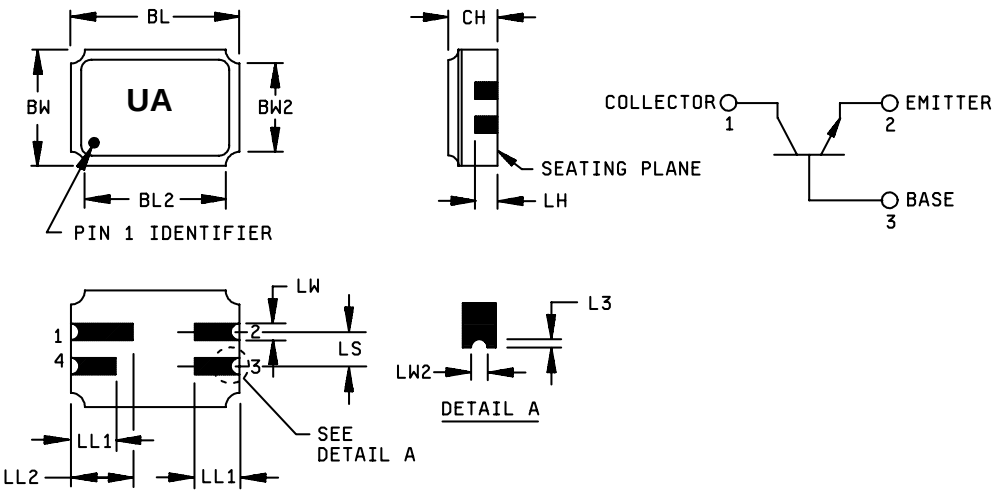


1.	Chip size.....	.040 x .040 inch ±.002 inch (1.02 mm x 1.02 mm ±0.05 mm).
2.	Chip thickness.....	.010 ± .0015 inch nominal (0.254 mm ±0.038 mm).
3.	Top metal.....	Aluminum 30,000Å minimum, 33,000Å nominal.
4.	Back metal.....	A. Al/Ti/Ni/Ag 12kÅ/3kÅ/7kÅ/7kÅminimum,15kÅ/ 5kÅ/10kÅ/10kÅ nominal. B. Gold 2.500Å minimum, 3000Å nominal.
5.	Backside.....	Collector.
6.	Bonding pad.....	B = .005 x .008 inch (0.127 mm x 0.203 mm). E = .010 x .007 inch (0.254 mm x 0.178 mm).

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions JANHCB and JANKCB (die) B versions.

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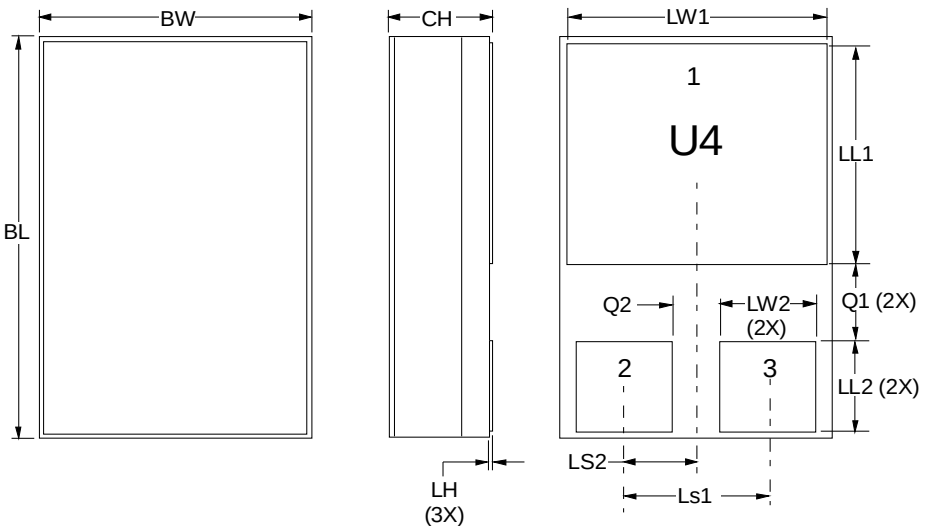
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
BL	.215	.225	5.46	5.71	
BL2		.225		5.71	
BW	.145	.155	3.68	3.93	
BW2		.155		3.93	
CH	.061	.075	1.55	1.90	3
L3	.003	.007	0.08	0.18	5
LH	.029	.042	0.74	1.04	
LL1	.032	.048	0.81	1.22	
LL2	.072	.088	1.83	2.23	
LS	.045	.055	1.14	1.39	
LW	.022	.028	0.56	0.71	
LW2	.006	.022	0.15	0.56	5

Pin no.	1	2	3	4
Transistor	Collector	Emitter	Base	N/C

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension "CH" controls the overall package thickness. When a window lid is used, dimension "CH" must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions " LW2" minimum and "L3" minimum and the appropriate castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension " LW2" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.
6. The coplanarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15mm) for solder dipped leadless chip carriers.

* FIGURE 4. Physical dimensions, surface mount (2N3439UA, 2N3440UA) version.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.72
BW	.145	.155	3.68	3.94
CH	0.049	0.075	1.24	1.91
LH		0.005		0.125
LW1	.135	.145	3.43	3.68
LW2	.047	.057	1.19	1.45
LL1	0.115	0.125	2.91	3.17
LL2	0.045	0.055	1.14	1.64
LS1	0.070	0.095	1.48	2.41
LS2	0.038	0.048	0.825	1.21
Q1	.045	.070	1.14	1.78
Q2	0.025	0.035	0.635	0.88
1	Collector			
2	Base			
3	Emitter			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Terminal 1 is collector.
4. Terminal 2 is base.
5. Terminal 3 is emitter.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 5. Physical dimensions, surface mount (2N3439U4, 2N3440U4) version (U4).

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

$R_{\theta JA}$	Thermal resistance junction to ambient.
$R_{\theta JC}$	Thermal resistance junction to case.
$R_{\theta JSP}$	Thermal resistance junction to solder pads.
UA	Surface mount case outlines (see figure 4).
TRB	Technical review board.
T_{SP}	Temperature of solder pads.

* 3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (similar to TO- 5 and TO-39), figure 2 (JANHCA and JANKCA, (A versions)), figure 3 (JANHCB and JANKCB (B versions)), figure 4 (2N3439UA and 2N3440UA surface mount, and figure 5 (2N3439U4 and 2N3440U4)).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking may be omitted from the body, but shall be retained on the initial container.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and table II herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3)	Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.3)
7	Hermetic seal (optional)	Optional
9	I_{CBO1} and h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1}; h_{FE1}; ΔI_{CB01} = 100 percent of initial value or 0.5 µA dc, whichever is greater; Δh_{FE1} = ± 15 percent of initial value.	I_{CBO1} and h_{FE1}
12	See 4.3.2	See 4.3.2
13	Subgroups 2 and 3 of table I herein; ΔI_{CB01} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE1} = ± 15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CB01} = 100 percent of initial value or 200 nA dc, whichever is greater; Δh_{FE1} = ± 15 percent of initial value.
14	Required	Required

4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500. As a minimum, die shall be 100 percent probed to ensure the assembled chips will meet the requirements of table I, subgroup 2.

* 4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. This option is limited to plants who are at least transitional (QML) approved or have an approved technical review board (TRB).

* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit used in screen 3c of 4.3, shall comply with the thermal impedance graph in figures 10, 11, 12, and 13 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3131.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in VIa (JANS) of 4.4.2.1 herein. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.3 herein. See 4.4.2.2 and table VIb of MIL-PRF-19500 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.3 herein.

* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc, 2,000 cycles.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hrs minimum, sample size in accordance with table VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hrs minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve T_J = +225°C minimum.
B6	3131	$R_{\theta JA}$ for TO-5 and TO-39, $R_{\theta JC}$ for U4 only (see 1.3 and 4.5.2).

* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a group B failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and /or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours, V_{CB} = 10 dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, VCB = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.3 herein.

* 4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E, except the UA package.
C5	3131	$R_{\theta JA}$ for TO-5 and TO-39, $R_{\theta JC}$ for U4 only, as applicable (see 1.3 and see 4.5.2).
C6	1026	1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

* 4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3). See 4.5.2.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with the steps of 4.5.3.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

* 4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following conditions shall apply:

a. Collector current magnitude during power application shall be 150 mA dc for $R_{\theta JC}$.

b. Collector emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be $+25°C \leq T_R \leq +35°C$. The chosen reference temperature shall be recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case for $R_{\theta JC}$.

f. Maximum limit see figures 10, 11, 12, and 13.

* 4.5.3 Delta requirements. Delta requirements shall be as specified below: (1) (2) (3) (4) (5)

Step	Inspection	MIL-STD-750		Symbol	Unit
		Method	Conditions		
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 360 V dc for 2N3439, 2N3439L, 2N3439UA V_{CB} = 250 V dc for 2N3440, 2N3440L, 2N3440UA	ΔI_{CB01} (1)	100 percent of initial value or 200 nA dc, whichever is greater.
2	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 20 mA dc; pulsed see 4.5.1	Δh_{FE1} (1)	±15 percent change from initial reading.

(1) Devices which exceed the table I limits for this test shall not be accepted.
(2) The delta measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:
  a. B4, see 4.5.3 herein, steps 1 and 2.
  b. B5, see 4.5.3 herein, steps 1 and 2.
(3) The delta measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows: Step 1, see 4.5.3 herein, steps 1 and 2.
(4) The delta measurements for appendix E, table VII (JANS) of MIL-PRF-19500 are as follows: C6, see 4.5.3 herein, steps 1 and 2.
(5) The delta measurements for appendix E, table IX of MIL-PRF-19500 are as follows:
  a. E1, see 4.5.3, steps 1 and 2.
  b. E2, see 4.5.3, steps 1 and 2.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
		Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to 3/ 4/ 5/ solvent	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 4/	1071	n = 22 devices, c = 0				
Fine leak Gross leak						
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 7 V dc	I_{EBO1}		10	µA dc
Collector to emitter cutoff	3041	Bias condition D	I_{CEO}		2	µA dc
2N3439, 2N3439L, 2N3439UA 2N3440, 2N3440L, 2N3440UA		V_{CE} = 300 V dc V_{CE} = 200 V dc				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX}		5	µA dc
2N3439, 2N3439L, 2N3439UA 2N3440, 2N3440L, 2N3440UA		V_{CE} = 450 V dc V_{CE} = 300 V dc				
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}		2	µA dc
2N3439, 2N3439L, 2N3439UA 2N3440, 2N3440L, 2N3440UA		V_{CB} = 360 V dc V_{CB} = 250 V dc				

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}		5	µA dc
2N3439, 2N3439L, 2N3439UA 2N3440, 2N3440L, 2N3440UA		V_{CB} = 450 V dc V_{CB} = 300 V dc				
Base emitter voltage (nonsaturated)	3066	Test condition A, pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{BE(sat)}$		1.3	V dc
Collector to emitter voltage (saturated)	3071	Pulsed (see 4.5.1), I_C = 50 mA dc, I_B = 4 mA dc	$V_{CE(sat)}$		0.5	V dc
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 20 mA	h_{FE1}	40	160	
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 2 mA	h_{FE2}	30		
Forward-current transfer ratio	3076	Pulsed (see 4.5.1), V_{CE} = 10 V dc, I_C = 0.2 mA	h_{FE3}	10		
Subgroup 3		T_A = +150°C				
High temperature operation:						
Collector to emitter cutoff current	3036	Bias condition D	I_{CB03}		6	µA dc
2N3439, 2N3439L, 2N3439UA 2N3440, 2N3440L, 2N3440UA		V_{CB} = 360 V dc V_{CB} = 250 V dc				
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 20 mA dc, pulsed (see 4.5.1)	h_{FE4}	15		
Subgroup 4						
Pulse response:	3251	Test condition A				
Turn-on time		V_{CC} = 200 V dc, I_C = 20 mA dc, I_{B1} = 2 mA dc, see figure 14	t_{on}		1	µs
Turn-off time		V_{CC} = 200 V dc, I_C = 20 mA dc, I_{B1} = -I_{B2} = 2 mA dc, see figure 14	t_{off}		10	µs

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Magnitude of common- emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc, I_C = 10 mA dc, f = 5 MHz	$\|h_{fe}\|$	3	15	
Open capacitance input open circuited	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MH	C_{obo}		10	pF
Small-signal short- circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 5 mA, f = 1 kHz	h_{fe}	25		
Input capacitance (output open circuited)	3240	V_{CB} = 5 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		75	
Subgroup 5						
Safe operating area (continuous dc)	3051	(See figure 15) T_C = +25°C, 1 cycle, t = 1.0 s.				
Test 1		V_{CE} = 5 V dc, I_C = 1 A dc				
Test 2						
Only 2N3439, 2N3439L, 2N3439UA		V_{CE} = 350 V dc, I_C = 14 mA dc				
Test 3						
Only 2N3440, 2N3440L, 2N3440UA		V_{CE} = 250 V dc, I_C = 20 mA dc				
Electrical measurements		See table I, subgroup 2 and 4.5.3 herein.				
Breakdown voltage, collector to emitter	3011	I_C = 10 mA, R_{BB1} = 470 ohms V_{BB1} = 6 V, L = 25 mH minimum f = 30 to 60 Hz	$V_{BR(CEO)}$			V dc
2N3439, 2N3439L, 2N3439UA 2N3440, 2N3440L, 2N3440UA				350 250		

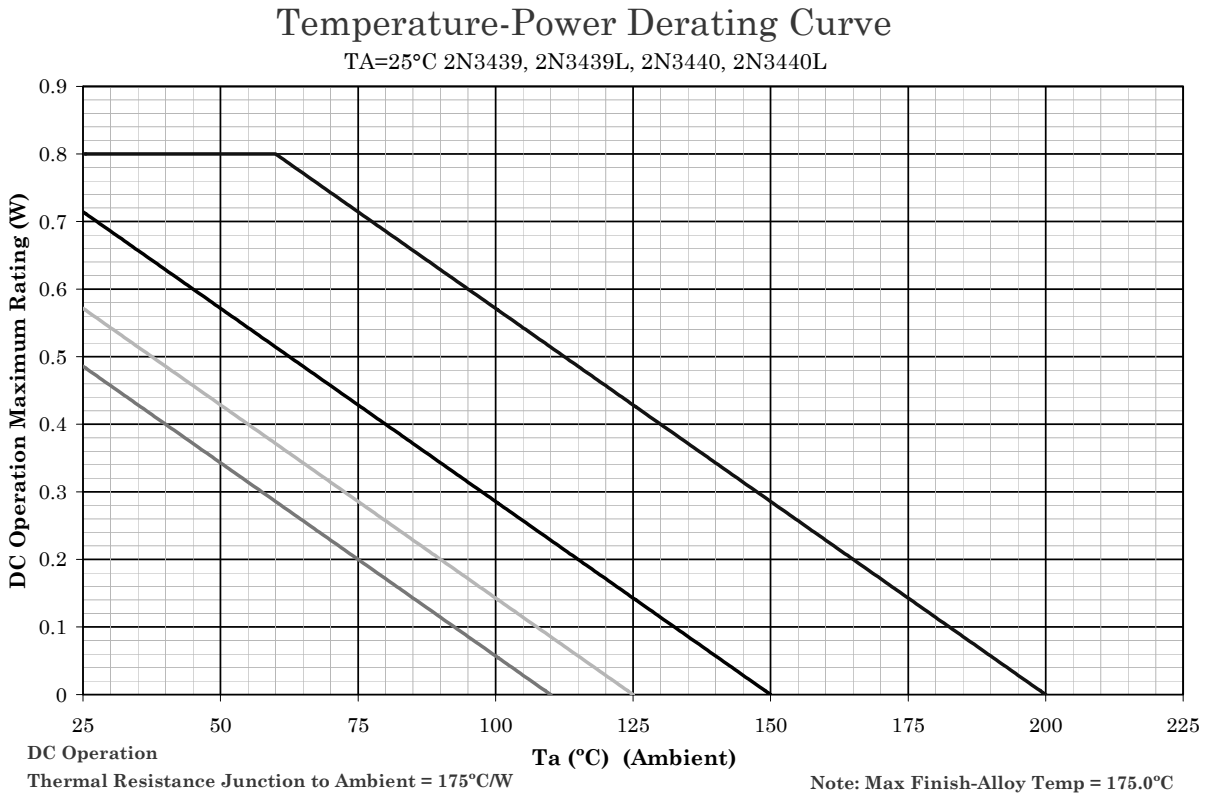
1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1 of table I, double the sample size of the failed test or sequence of tests.
3/ Separate samples may be used.
4/ Not required for JANS.
5/ Not required for laser marked devices.

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TABLE II. Group E inspection (all quality levels) - for qualification only or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.3.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: $V_{CB} \geq 10$ V dc, 6,000 cycles	
Electrical measurements		See table I, subgroup 2 and 4.5.3.	
Subgroup 4			
Thermal resistance	3131	$R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets).	15 devices, c = 0
Thermal impedance curves		Each supplier shall submit their qualification lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	sample size N/A
Subgroup 5			15 devices, c = 0
Barometric pressure (reduced)	1001	Normal mounting pressure = 8 mm Hg ±2 mm Hg for 60 s (minimum)	
Electrical measurements		See table I, subgroup 2 and 4.5.3.	
Subgroups 6			3 devices c = 0
ESD (electrocstatic discharge)	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V, condition B for devices < 400 V.	

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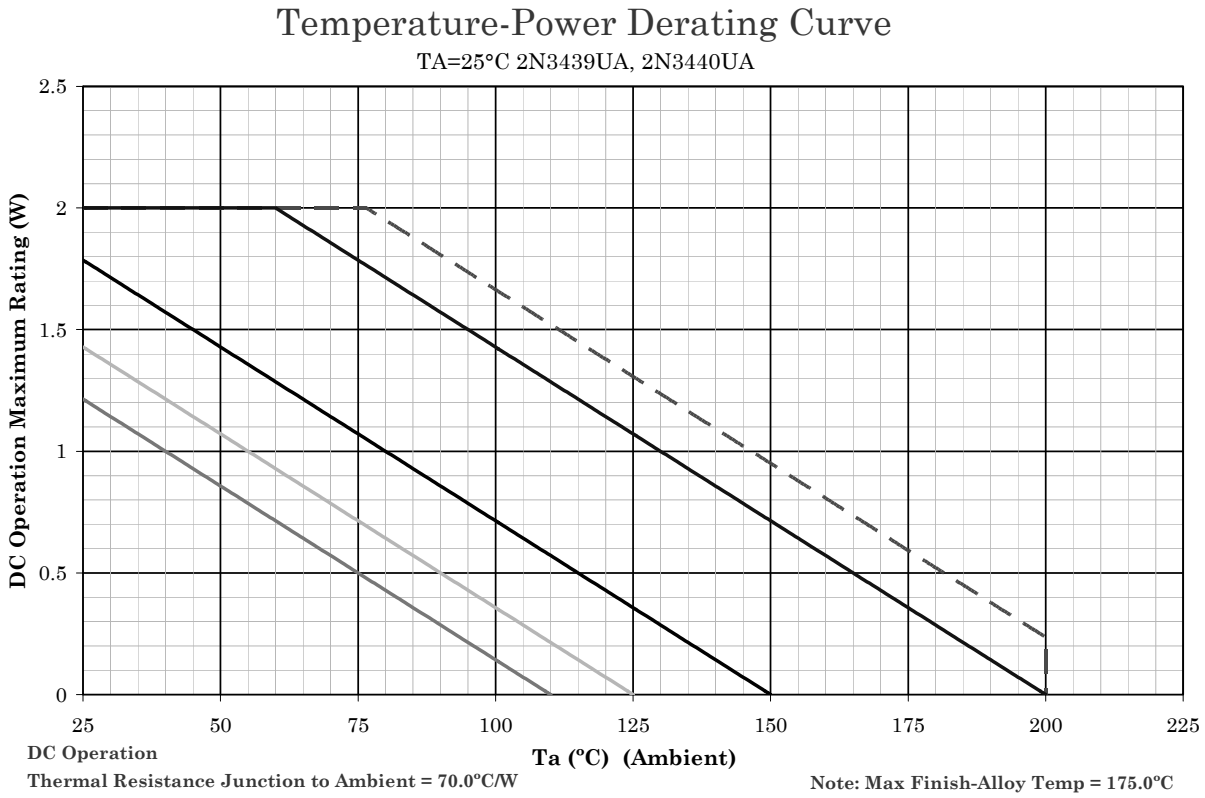


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 6. Temperature-power derating for 2N3439, 2N3439L, 2N3440, 2N3440L, $R_{\theta JA}$ (TO-5 and TO-39).

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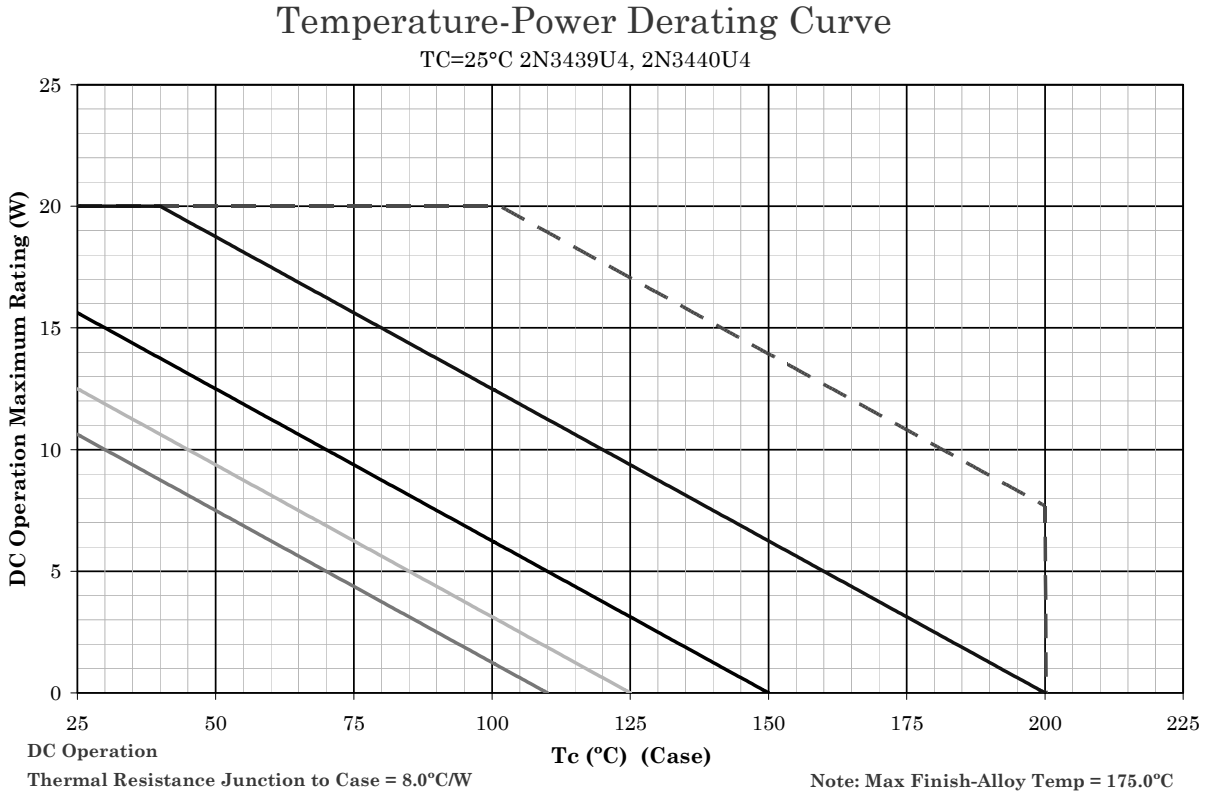


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 7. Temperature-power derating for 2N3439UA, and 2N3440UA.

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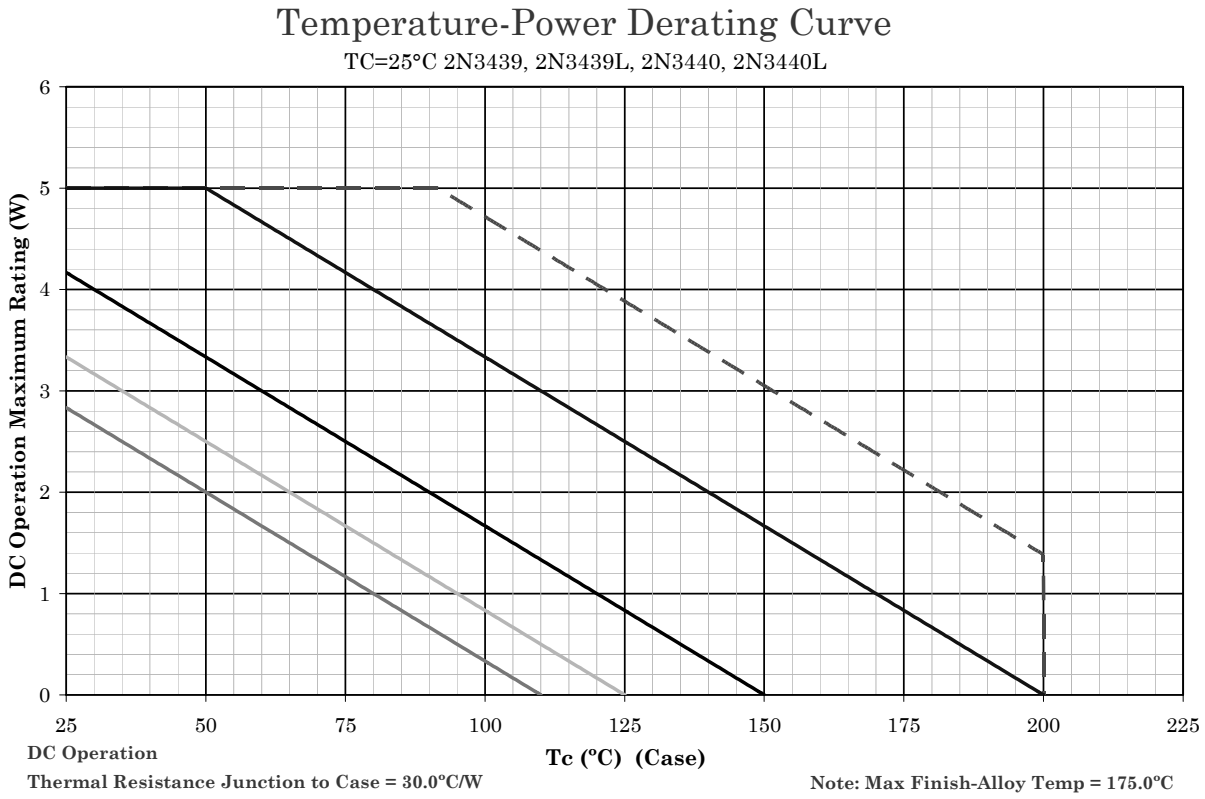


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 8. Temperature-power derating for 2N3439U4, and 2N3440U4.

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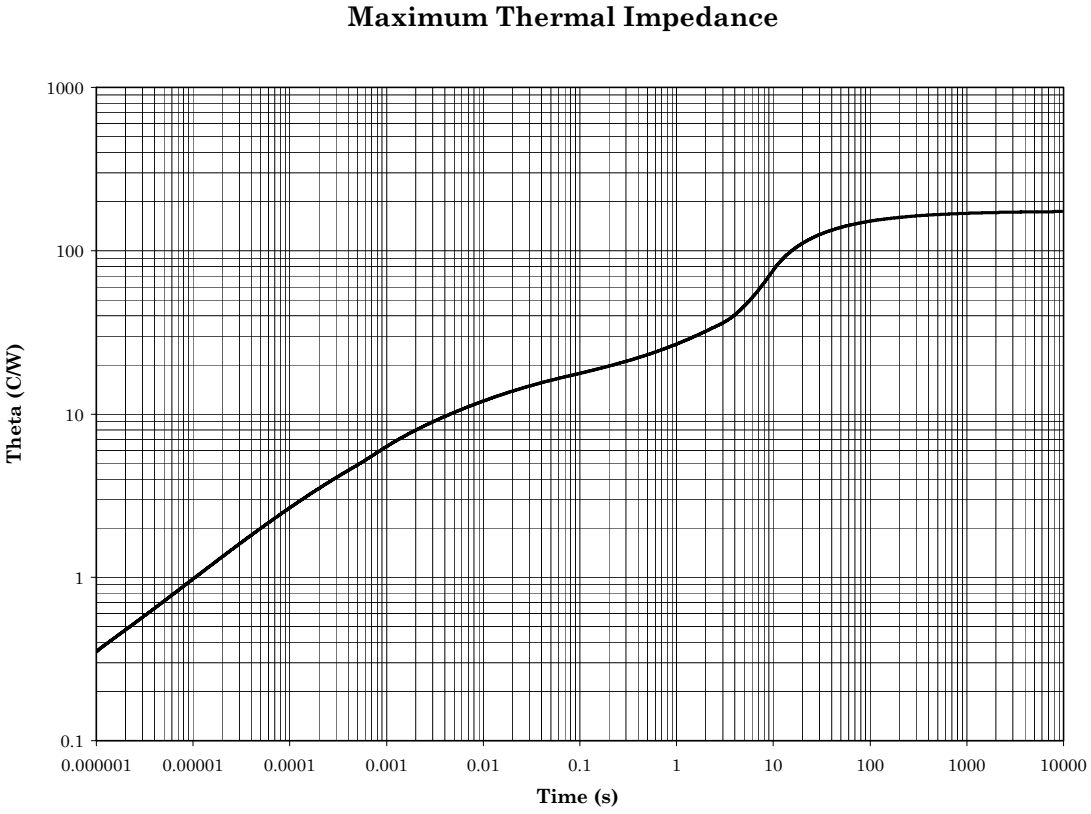


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 9. Temperature-power derating for 2N3439, 2N3439L, 2N3440 and 2N3440L.

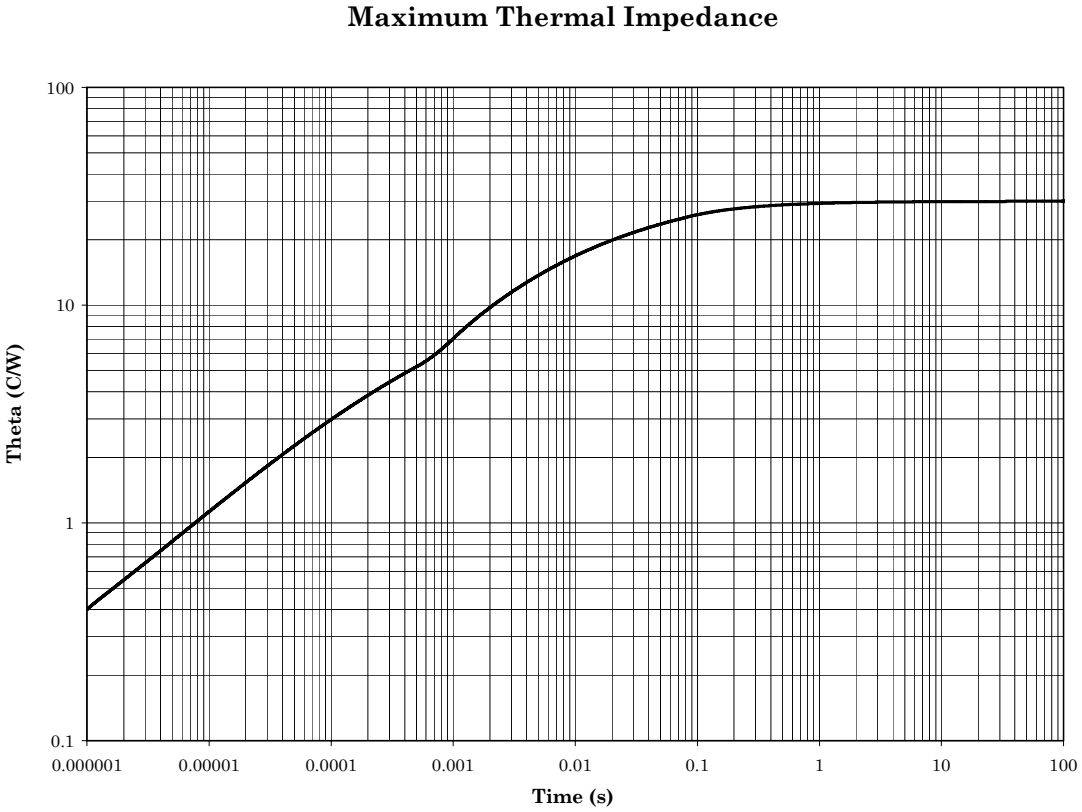
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T_A = +25°C, P_T = 0.8W, thermal resistance $R_{\theta JA}$ = 175°C/W.

* FIGURE 10. Thermal impedance graph ($R_{\theta JA}$) for 2N3439, 2N3439L, 2N3440, and 2N3440L (TO-5, TO-39, and UA).

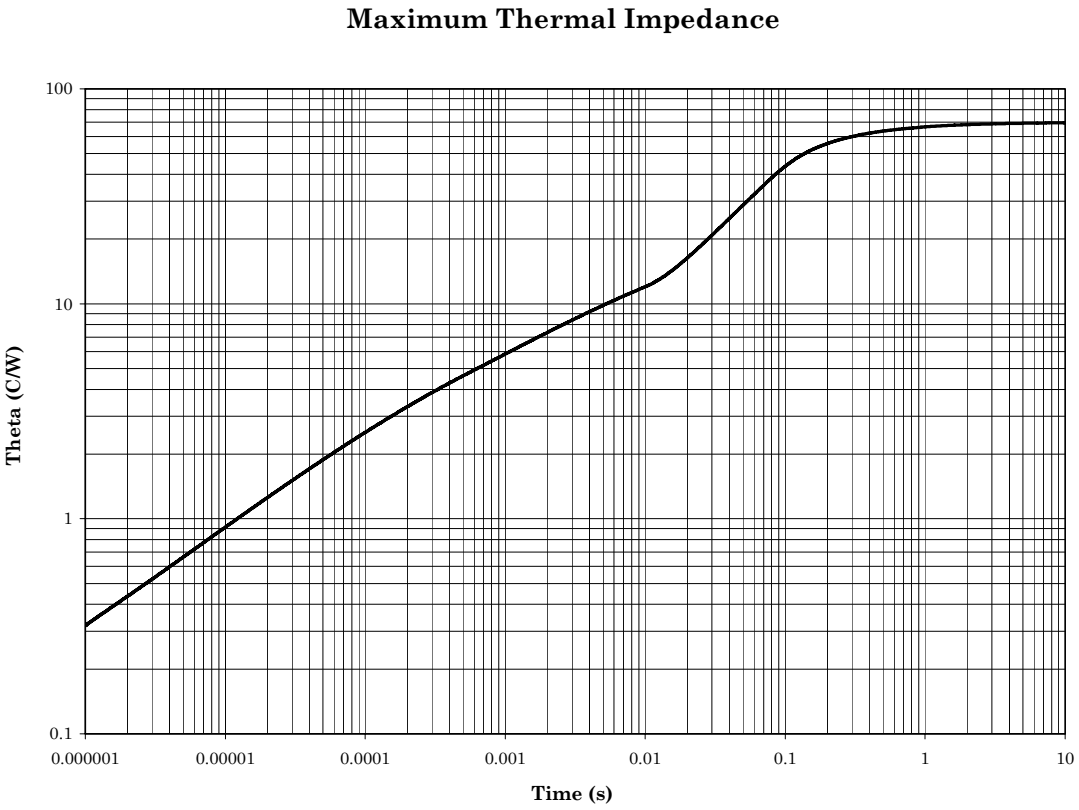
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T_C = +25°C, P_T = 5.0W, thermal resistance $R_{\theta JC}$ = 30°C/W, steel.

* FIGURE 11. Thermal impedance graph ($R_{\theta JC}$) for 2N3439, 2N3439L, 2N3440, and 2N3440L (TO-5 and TO-39).

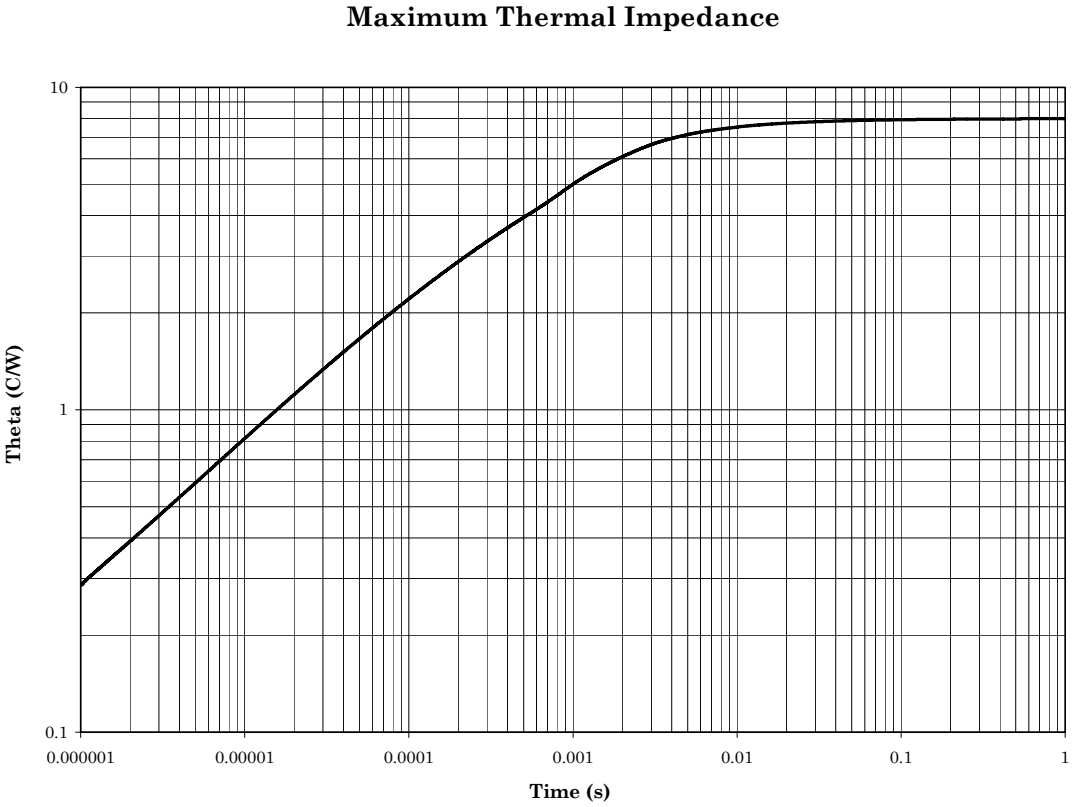
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T_C = +25°C, thermal resistance $R_{\theta JC}$ = 70°C/W, Pdiss = 2W.

* FIGURE 12. Thermal impedance graph ($R_{\theta JC}$) for 2N3439, 2N3439L, 2N3440, and 2N3440L (UA).

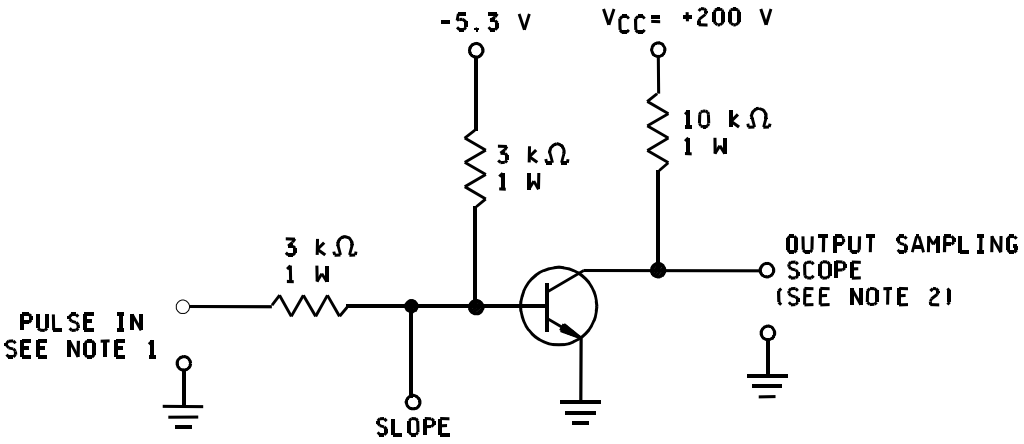
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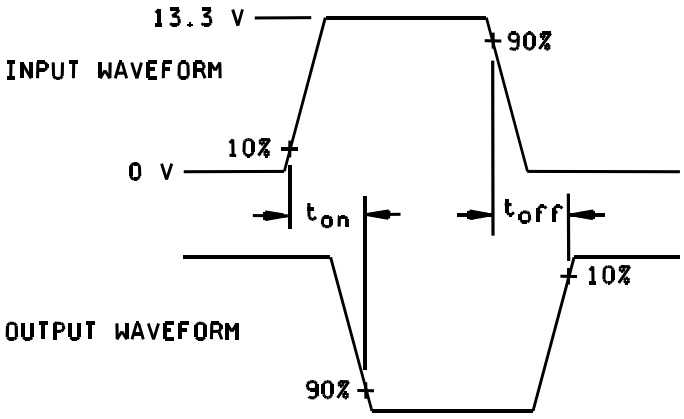


Solder mounted to copper heatsink at T_C = +25C. Thermal resistance $R_{\theta JC}$ = 8.0 C/W

* FIGURE 13. Thermal impedance graph ($R_{\theta JC}$) for 2N3439U4, and 2N3440U4 (U4).

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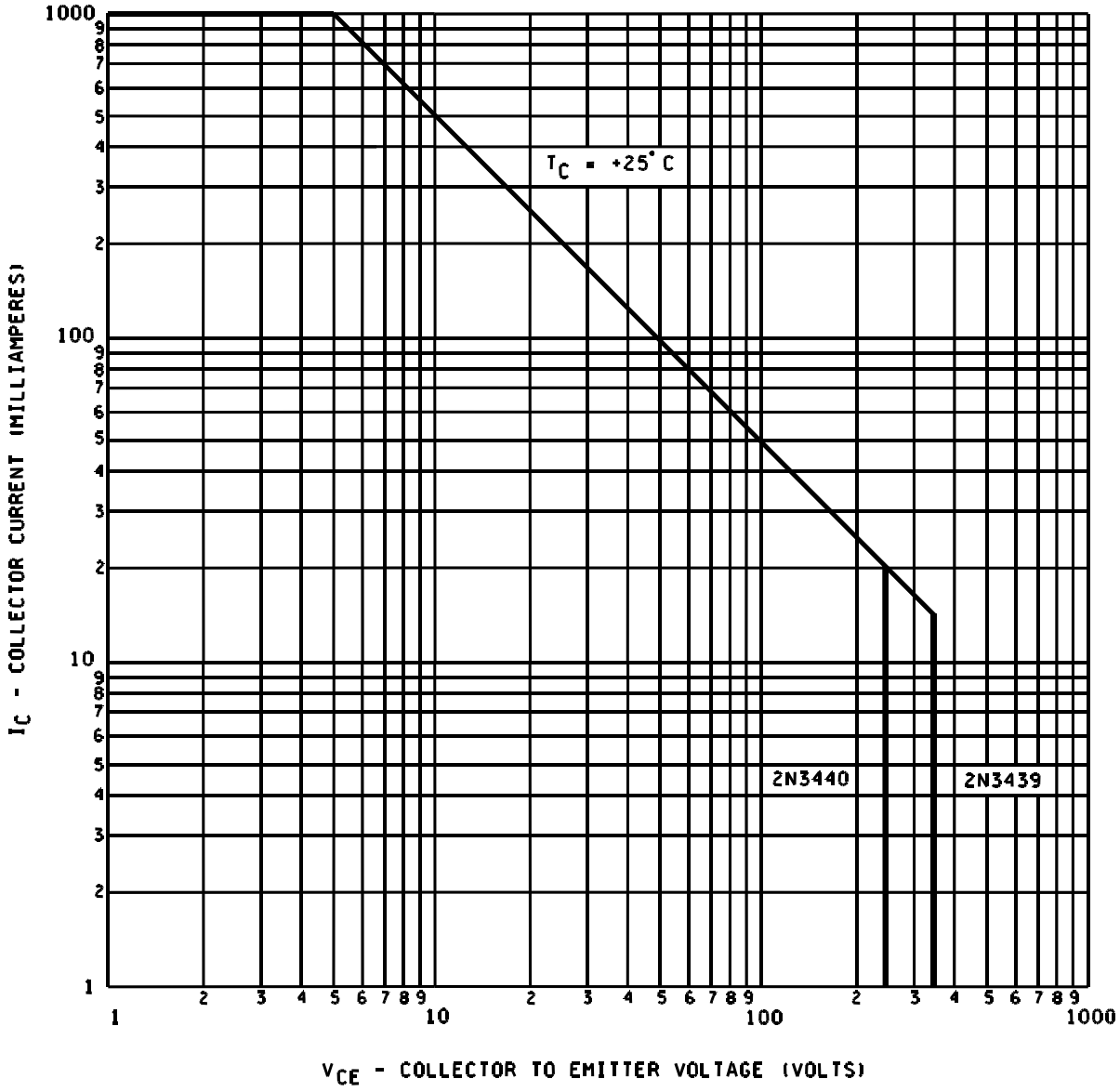


NOTES:

1. The rise time (t_r) and fall time (t_f) of applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, generator source impedance shall be 50 ohms, pulse width = 20 μs.
2. Output sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 50 pF, and rise time ≤ 1.0 μs.

* FIGURE 14. Pulse response test circuit.

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NOTE: Also applies to the corresponding "L" and "UA" suffix devices.

* FIGURE 15. Maximum safe operating graph (continuous dc).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3439) will be identified on the QML.

Die ordering information		
PIN	Manufacturer	
	33178	43611
2N3439	JANHCA2N3439 JANKCA2N3439	JANHCB2N3439 JANKCB2N3439
2N3440	JANHCA2N3440 JANKCA2N3440	JANHCB2N3440 JANKCB2N3440

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2709)

Review activities:
Army - AR, MI, SM
Navy - AS, MC
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